

Shahkar Ahmad

AI Engineer / Agentic Systems, LLM Applications & Applied ML

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SUMMARY

Final-year Computer Science student (AI specialisation) who builds agentic AI systems end to end — LangGraph state machines, hybrid retrieval pipelines, and the evaluation harnesses that prove they work rather than merely demo well. Recent work includes a voice-first multi-agent moot-court simulator whose retrieval scores **MRR 1.00** with **100% citation accuracy** against an authenticated legal corpus, and a 13-agent content engine with human-in-the-loop control and LLM-as-judge scoring. Comfortable across the stack: Python and FastAPI for reasoning services, TypeScript and React for the surface, PostgreSQL underneath. Seeking an AI/ML internship or a remote AI Engineer role.

TECHNICAL SKILLS

Agentic AI	LangGraph · LangChain · Multi-Agent Orchestration · ReAct & Supervisor Patterns · Human-in-the-Loop · Tool Calling · MCP
RAG & Retrieval	Hybrid BM25 + dense vectors · Reciprocal Rank Fusion · LLM reranking · Semantic chunking · Citation grounding & auditing
LLMs & Evaluation	GPT-4o / 4o-mini · Gemini 2.5 Flash (native audio) · OpenAI Embeddings · Prompt Engineering · DeepEval G-Eval · LLM-as-Judge · recall@k / MRR / F1
Machine Learning	Scikit-learn · TensorFlow / Keras · CNN, RNN, MLP · Pandas · NumPy · Matplotlib / Seaborn · EDA · Feature Engineering
Backend & Data	Python · FastAPI · Express.js · TypeScript · PostgreSQL · Drizzle ORM · Supabase · SQLite · OpenAPI / REST · WebSockets
Frontend & Tools	React 19 · Vite · Streamlit · Tailwind CSS · Git & GitHub · Docker · Jupyter · Power BI
Languages	English · Urdu · Pashto

SELECTED PROJECTS

Adalat AI — Voice-First AI Moot-Court Simulator [Shah-123/CourtSimulator](#) · 2026

- Architected a **LangGraph state machine** where a supervisor routes each student utterance through independent Judge, Opposing Counsel and Witness agents, backed by two-tier memory — verbatim working memory plus a summarised case file — so counsel catches a student who changes their story across phases.
- Built a **ReAct Judge** that calls a `search_statute` tool to read the governing provision before ruling, and an Opposing Counsel raising autonomous evidentiary objections verified against the Qanun-e-Shahadat Order 1984.
- Engineered a hybrid **RAG pipeline** fusing BM25 and dense vectors via Reciprocal Rank Fusion ($k = 60$) with LLM reranking, plus a citation audit engine that flags fabricated provisions rather than presenting them as settled law.
- Designed a contract-first microservice architecture — Express API, Python FastAPI reasoning service, OpenAPI-generated clients and validators — with streaming STT/TTS, cutting time-to-first-audio from **16.1s to 6.9s**.
- Established an evaluation harness over **32 labelled scenarios**: retrieval MRR 1.00, objection F1 1.00, citation accuracy 100% versus 0% for the ungrounded baseline.

Tech: LangGraph · Python · FastAPI · Express.js · React 19 · OpenAI API · PostgreSQL · Drizzle ORM

AI Content Factory — Multi-Agent Content Engine [FINAL YEAR PROJECT](#) [Shah-123/AI-Content-Factory](#) · 2025 – 2026

- Built a **13-agent LangGraph pipeline** — router, researcher, orchestrator, parallel workers, QC auditor, SEO optimiser, podcast studio, video generator and academic judge — turning one prompt into a researched blog post, social campaign, podcast and short-form video.
- Parallelised the writing phase with fan-out worker agents scoped to their own evidence records, cutting generation time against sequential writing while preventing redundant text and source-stuffing.
- Implemented **Human-in-the-Loop** outline control via LangGraph interrupts and a SQLite checkpoint, so users approve or rewrite the plan before costly generation and interrupted runs resume from the last saved step instead of repeating LLM calls.
- Grounded output in retrieved evidence — semantic chunking over `text-embedding-3-small` plus a Tavily-backed fact-checking auditor — and scored every run with an in-house rubric and **DeepEval G-Eval** across coherence, relevance, grounding and tone.

Tech: LangGraph · Python 3.11 · FastAPI · OpenAI API · Gemini 2.5 Flash · Tavily · DeepEval · MoviePy · React

Live Agent Hub — AI Streamer Practice Platform

live-agent-hub.vercel.app · 2025

- Built a full-stack platform that simulates a live Twitch chat with **8 distinct AI personas** driven by OpenAI completion endpoints, letting streamers rehearse against a realistic audience.
- Engineered humanised agent traits — slang, typo simulation, chattiness filters and dynamic reaction to on-screen context — so the room reads as people rather than a bot loop.
- Architected an Express.js backend on PostgreSQL/Supabase behind a React + Vite frontend, deployed and publicly playable.

Tech: React · Vite · TypeScript · Tailwind CSS · Radix UI · Express.js · PostgreSQL · Supabase · OpenAI API

ChessMastermind — Chess Engine Built From Scratch

[Shah-123/chessAgent](https://shah-123.github.io/chessAgent) · 2025

- Implemented minimax with **alpha-beta pruning**, iterative deepening, aspiration windows and quiescence search — no engine library doing the thinking.
- Optimised search with transposition tables (FEN hashing), MVV-LVA move ordering, killer-move and history heuristics.
- Wrote evaluation covering piece-square tables, pawn structure (passed/isolated), king safety and knight outposts, served through a FastAPI + Streamlit interface that explains its reasoning.

Tech: Python · python-chess · FastAPI · Streamlit · NumPy

EDUCATION

University of Peshawar — BS Computer Science, AI Specialisation

Peshawar, Pakistan · Oct 2022 – Oct 2026 (expected)

Final Year Project: AI Content Factory — stateful multi-agent content orchestration and evaluation engine. Relevant coursework:

Machine Learning · Deep Learning · Data Structures · Database Systems · Software Engineering.

ADDITIONAL ML & DATA WORK

- **Corrective RAG** — retrieval pipeline that grades the context it retrieves and re-queries when quality falls short, rather than answering from a weak match. [GitHub](#)
- **Polio Campaign Monitor** (2024) — forecast “Not Available Children” field metrics to help supervisors allocate vaccination resources; ~91% accuracy on messy real-world data, delivered as a Streamlit dashboard.
- **ODI Run Predictor** — regression model (R^2 0.92, MSE 0.11) predicting cricket innings totals from match conditions, exposed via a FastAPI endpoint and a Streamlit UI. [GitHub](#)
- **YouTube Video Summariser** — LangChain pipeline turning transcripts into chapter summaries, key insights and action items. [GitHub](#)
- **Also:** Heart Failure Prediction (clinical risk classification) · Google Play Store EDA across 10,000+ apps · deep-learning mini-projects in CNNs, RNNs and transfer learning · Power BI KPI dashboards.

Open to AI/ML internships, remote AI Engineer roles and freelance LLM & agentic-AI contracts · UTC+5, flexible hours · References available on request